

Customer Segmentation Using K-Means and Hierarchical Clustering

Machine Learning (25SC2107E)

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ABSTRACT

Customer segmentation is a key technique used by businesses to understand customer behavior and improve marketing strategies. By dividing customers into different groups based on their characteristics and purchasing patterns, companies can provide personalized services and make better business decisions. This project focuses on customer segmentation using two popular unsupervised machine learning algorithms: K-Means Clustering and Hierarchical Clustering. The project uses a customer dataset containing demographic and purchasing information. Before applying the clustering algorithms, the data is preprocessed by handling missing values, encoding categorical data, and performing feature scaling to improve the accuracy of the models. The K-Means algorithm groups customers into different clusters, while the optimal number of clusters is determined using the Elbow Method and Silhouette Analysis. Hierarchical Clustering is then applied to visualize the relationships between customer groups using a dendrogram, allowing a comparison of its performance with K-Means Clustering.

The generated clusters help identify different customer segments, such as highvalue customers, regular customers, and customers who may require additional attention. These insights enable businesses to design targeted marketing campaigns, improve customer satisfaction, enhance customer retention, and increase overall profitability. The comparison of both clustering techniques also highlights their strengths and limitations in customer segmentation. Overall, this project demonstrates how machine learning can be effectively used to analyze customer data and support data-driven business decision

Course Instructor

Signature of the Guide